

Adaptive multi-robot coordination for heterogeneous quadrupeds: a literature review

No published work exists on learned coordination between Boston Dynamics Spot and Ghost Robotics Vision 60, making this project a genuine first-mover contribution at the intersection of heterogeneous MARL, quadruped sim-to-real transfer, and physical multi-robot deployment. The algorithmic foundations are mature — MAPPO, HAPPO, and capability-conditioned architectures provide strong starting points — but the bridge between multi-agent RL and real quadruped hardware remains almost entirely unbuilt. Recent 2024–2026 work on single-policy multi-morphology locomotion (ManyQuadrupeds), decentralized multi-quadruped transport (decPLM), and the Spot RL Researcher Kit dramatically lower the barriers to entry. The DARPA Subterranean Challenge produced the most relevant systems-level references, with teams like CERBERUS, CoSTAR, and CTU-CRAS-NORLAB fielding heterogeneous fleets including Spot quadrupeds in GPS-denied environments. (MDPI +2)

This review consolidates **75+ papers and codebases** across three focus areas, prioritizing open-source implementations and actionable architectural choices for a shared PPO policy with capability-conditioned observations in a PettingZoo-style simulation.

1. Core MARL methods for heterogeneous teams

Foundational algorithms that define the landscape

VDN (Sunehag et al., AAMAS 2018; arXiv:1706.05296) decomposes the team value function into an additive sum of per-agent utilities. QMIX (Rashid et al., ICML 2018; arXiv:1803.11485) relaxes this to monotonic mixing via hypernetworks conditioned on global state,

(George Washington Univer...) becoming the dominant value-decomposition baseline on SMAC.

(APXML +3) Both are implemented in PyMARL (github.com/oxwhirl/pymarl). **MADDPG** (Lowe et al., NeurIPS 2017; arXiv:1706.02275) introduced centralized-critic decentralized-actor training for continuous actions, (GitHub) directly relevant to robot control; (PubMed Central) code at github.com/openai/maddpg. **COMA** (Foerster et al., AAI 2018; arXiv:1705.08926) addresses credit assignment through counterfactual baselines — marginalizing out each agent's action while fixing others — computable in a single critic forward pass. (ResearchGate) (ACM Digital Library)

The watershed result for this project is **MAPPO** (Yu et al., NeurIPS 2022; arXiv:2103.01955), demonstrating that PPO with a centralized value function matches or exceeds off-policy methods across SMAC, MPE, Hanabi, and Google Research Football. (arXiv) Five implementation details proved critical: **PopArt value normalization**, proper clipping ranges, limited training epochs per batch, death masking, and centralized-state value inputs. The official codebase (github.com/marlbenchmark/on-policy) is the natural starting point for the project's shared PPO

architecture. (GitHub) IPPO (de Witt et al., 2020; arXiv:2011.09533) — independent PPO without centralized information — serves as the essential ablation baseline, confirming that even naive independence produces surprisingly competitive results in some cooperative settings.

Heterogeneous-agent algorithms from 2022 onward

HAPPO and HATRPO (Kuba et al., ICLR 2022; arXiv:2109.11251; (arXiv) extended in Zhong et al., JMLR 2024; arXiv:2304.09870) provide the first trust-region MARL methods with **monotonic improvement guarantees for heterogeneous agents**. (arXiv) The key innovation is a multi-agent advantage decomposition lemma enabling sequential policy updates (arXiv) where each agent's advantage is reweighted by importance-sampling ratios from preceding updates. (JMLR) Unlike MAPPO, HAPPO requires no parameter sharing and natively supports different observation/action spaces. (arXiv +2) The JMLR 2024 paper extends this into the broader HARL framework (HAA2C, HADDPG, HATD3, HASAC), (arXiv) with HASAC (arXiv:2306.10715) currently achieving state-of-the-art on continuous cooperative benchmarks. (arXiv) Code for the full suite: github.com/PKU-MARL/HARL. (GitHub)

MAT (Wen et al., NeurIPS 2022; arXiv:2205.14953) reframes cooperative MARL as sequence modeling: an encoder processes agents' observations via self-attention, and a decoder auto-regressively generates actions (arXiv) using the same advantage decomposition from HAPPO. (NeurIPS) MAT achieves superior data efficiency on SMAC, MaMuJoCo, and dexterous hands. Code: github.com/PKU-MARL/Multi-Agent-Transformer. Its successor **Sable** (Mahjoub et al., ICML 2025; arXiv:2410.01706) replaces attention with retention mechanisms for linear scaling (arXiv) to 1,000+ agents while adding temporal episode memory — implemented in the Mava framework (github.com/instadeepai/Mava). (arXiv) (InstaDeep)

SHPPPO (Guo et al., Neurocomputing 2025; arXiv:2404.03869) directly addresses the same design challenge as this project: combining parameter sharing with heterogeneous behavior. A LatentNet generates agent-specific distributions that parameterize a single heterogeneous layer in the actor network while all other parameters remain shared. (arXiv) SHPPPO outperforms both MAPPO and HAPPO on SMAC and GRF with zero-shot scalability to unseen team sizes. (arXiv) (The Moonlight)

Architectures for capability-conditioned heterogeneous teams

The most directly relevant architectural paper is **Torbati et al.** (CoRL 2024; arXiv:2401.13127), which develops policies with four properties: capability awareness (actions conditioned on continuous capability vectors), capability communication (graph networks share capability information across agents), robot-agnostic parameter sharing, and decentralized execution. (arXiv) Experiments demonstrate that **reasoning about capabilities rather than agent IDs is essential for generalization** to new team compositions and robot types. (arXiv) (OpenReview) This validates the project's core architectural choice.

HetGPPO (Bettini et al., CoRL 2023; arXiv:2301.07137) introduces a GNN-based heterogeneous PPO framework with differentiable inter-agent communication. The paper's key finding: shared-parameter policies achieve "apparent heterogeneity" that is brittle under deployment noise, while genuine heterogeneity via HetGPPO proves more resilient. Validated on real robots. (arXiv) (ResearchGate) Code: github.com/proroklab/HetGPPO.

UPDeT (Hu et al., ICLR 2021 Spotlight; arXiv:2101.08001) uses Transformer attention to decouple policy distributions from input observations, enabling a single model to handle agents with different observation/action configurations. (ADS +2) Code: github.com/Theohhhu/UPDeT. For GNN-based coordination, **MAGEC** (Goeckner et al., IROS 2024; arXiv:2403.13093) combines GNN message-passing with MAPPO for resilient multi-robot coordination that handles agent attrition and communication disturbances. (PubMed Central) (arXiv)

Parameter sharing with agent-type conditioning is studied by Terry et al. (2020; arXiv:2005.13625) and Christianos et al. (ICML 2021), establishing that conditioned parameter sharing with agent-specific features achieves the benefits of both shared and independent learning. (PubMed Central) (arXiv) **Selective parameter sharing** — grouping agents by learned similarity — offers a middle ground.

Role emergence without hand-coded specialization

RODE (Wang et al., ICLR 2021; arXiv:2010.01523) discovers roles by clustering actions according to their environmental effects, then uses a bi-level hierarchy: a role selector at low temporal resolution and role-specific policies in reduced action spaces. Outperforms SOTA on 10/14 SMAC scenarios. Code: github.com/TonghanWang/RODE. **ROMA** (Wang et al., ICML 2020; arXiv:2003.08039) learns stochastic role embeddings with identifiability and specialization regularizers, producing dynamic role changes during episodes. (arXiv) (DeepAI) Code: github.com/TonghanWang/ROMA. Recent extensions include **Role Play** (Long et al., 2024), which recasts policy diversity as diversity in role embeddings using Social Value Orientation concepts — a common policy network accepts both observation and role embedding.

(Emergent Mind)

Reward shaping and credit assignment

Difference rewards (Wolpert & Tumer, 2002) provide agent-specific signals aligned with the global objective by computing each agent's marginal contribution (ResearchGate) — tractable for two agents. COMA's counterfactual baseline operationalizes this within actor-critic methods. (University of Oxford) **LIIR** (Du et al., NeurIPS 2019) learns parameterized intrinsic reward functions via bilevel optimization, naturally diversifying agent behaviors while maximizing team return. (arXiv) (NeurIPS) Code: github.com/yalidu/liir. **DAE** (Li et al., ICML 2022) provides a family of advantage estimators interpolating between individual and counterfactual advantages, with theoretical analysis showing the optimal mixing coefficient $\beta < 1$ due to self-consistency issues. (Proceedings of Machine Le...) Jiang et al. (Applied Intelligence, 2023) explicitly address credit

assignment for heterogeneous agents in cooperative tasks with continuous action spaces.

Springer

Curriculum learning, action masking, and partial observability

Huang & Ontañón (FLAIRS 2022; arXiv:2006.14171) provide the definitive treatment of **invalid action masking in PPO**: setting invalid logits to $-\infty$ before softmax produces valid policy gradient updates. Code: github.com/vwxyzjn/invalid-action-masking. This is critical for encoding which actions each quadruped type can execute in a shared policy.

For curriculum learning, Zhao et al. (ICLR 2025; arXiv:2205.10016) propose using **TD-error-based learning progress** rather than reward as the curriculum signal for MARL, which is more robust in cooperative settings. [OpenReview](#) [arXiv](#) A 2025 paper in Engineering Applications of Artificial Intelligence demonstrates curriculum learning for real UGV teams, progressing from single-agent navigation to coordinated multi-agent pathfinding with emergent task allocation.

ScienceDirect

For partial observability, DRQN (Hausknecht & Stone, 2015; arXiv:1507.06527) is foundational [Sage Journals](#) — nearly all SMAC methods use DRQN-style recurrent architectures. **AERIAL** (Phan et al., ICML 2023) replaces true-state access with learned joint memory representations using simplified transformers over agents' RNN hidden states, [Proceedings of Machine Le...](#) directly applicable to PettingZoo-based partially observable environments.

Communication mechanisms

CommNet (Sukhbaatar et al., NeurIPS 2016; arXiv:1605.07736; github.com/facebookresearch/CommNet) provides broadcast continuous communication layerable on shared PPO. TarMAC (Das et al., ICML 2019; arXiv:1810.11187; github.com/facebookresearch/tarmac) adds sender-receiver attention for targeted messaging. [arXiv](#) IC3Net (Singh et al., ICLR 2019; arXiv:1812.09755; github.com/IC3Net/IC3Net) adds learned gating for communication timing. [arXiv](#) For two agents, the communication overhead is minimal — CommNet or a simple message-passing layer is sufficient.

2. Sim-to-real transfer for quadruped robots

Domain randomization established on a Ghost Robotics platform

The foundational sim-to-real paper for quadruped RL is **Tan et al.** (RSS 2018; arXiv:1804.10332), which achieved zero-shot transfer of trotting and galloping gaits on the **Ghost Robotics Minitaur** using three techniques: system identification (accurate actuator model + latency simulation), dynamics randomization (mass, friction, motor parameters), and compact observation space design. [Google Research](#) [arXiv](#) A critical finding: **a 4D IMU-only observation transferred better than a 12D observation including motor angles.** [GitHub](#)

(Robotics: Science and Syst...) Code via PyBullet (github.com/bulletphysics/bullet3). Peng et al. (ICRA 2018) formalized dynamics randomization — randomizing mass, friction, and damping each episode — as a general technique. Muratore et al. (Frontiers in Robotics and AI, 2022) provide a comprehensive review of domain randomization approaches.

The ETH Zurich RSL pipeline: the field's gold standard

Hwangbo et al. (Science Robotics, 2019; DOI:10.1126/scirobotics.aau5872) introduced the **learned actuator network** for bridging the sim-to-real gap on ANYmal, achieving agile velocity tracking and fall recovery that surpassed model-based controllers. This actuator network approach remains core to the RSL pipeline. (Robotic Systems Lab) **Lee et al.** (Science Robotics, 2020) extended this with a teacher-student paradigm for blind (proprioception-only) locomotion over challenging terrain (mud, snow, rubble) via extensive domain randomization, achieving zero-shot generalization to natural environments never seen in training. (Robotic Systems Lab) Project: leggedrobotics.github.io/rl-blindloco.

Miki et al. (Science Robotics, 2022; arXiv:2201.08117) added exteroceptive perception, training a teacher with privileged terrain ground-truth while a student reconstructs environment state from noisy real-world height samples. Deployed zero-shot on ANYmal traversing stairs in any orientation. (Leggedrobotics) Project: leggedrobotics.github.io/rl-perceptiveloco.

Legged Gym (Rudin et al., CoRL 2022; arXiv:2109.11978) became the de facto training framework by enabling massively parallel training of locomotion policies on NVIDIA Isaac Gym — training ANYmal on rough terrain in minutes using 4,096+ parallel environments. (GitHub) Code: github.com/leggedrobotics/legged_gym. (GitHub) **RSL-RL** (Schwarke et al., 2025; arXiv:2509.10771; github.com/leggedrobotics/rsl_rl) is the successor training library, (arXiv) now compatible with Isaac Lab and MuJoCo Playground. (GitHub)

For system identification, **PACE** (Bjelonic et al., 2025; arXiv:2509.06342; github.com/leggedrobotics/pace-sim2real) combines evolutionary optimization with learned actuator dynamics for data-driven sim-to-real calibration, tested on ANYmal and other platforms. **SPI-Active** (Sobanbabu et al., CoRL 2025; arXiv:2505.14266; github.com/LeCAR-Lab/SPI-Active) uses pre-trained RL policies for active exploration that maximizes Fisher Information of trajectories — outperforming baselines by **42–63%** on Unitree Go2/G1, directly applicable when proprietary dynamics models are unavailable.

Rapid Motor Adaptation and agile locomotion

RMA (Kumar et al., 2021; arXiv:2107.04034) introduced a two-component architecture: a base policy conditioned on environment extrinsics plus an adaptation module that infers these extrinsics from proprioceptive history in real-time. Deployed on Unitree A1 across rocky, slippery, deformable, sandy, and grassy terrains without fine-tuning. Project: ashish-kmr.github.io/rma-legged-robots. **Margolis et al.** (RSS 2022; arXiv:2205.02824) achieved **3.9 m/s** on MIT Mini Cheetah using adaptive curriculum and online system identification. (arXiv) (arXiv)

Walk These Ways (Margolis & Agrawal, CoRL 2023; arXiv:2212.03238; github.com/Improbable-AI/walk-these-ways) trains a single policy encoding a structured family of locomotion strategies selectable at runtime (OpenReview) (arXiv) — directly useful when coordinating robots that need behavior-switching for different roles.

Extreme Parkour (Cheng et al., ICRA 2024; arXiv:2309.14341; github.com/chengxuxin/extreme-parkour) demonstrates highly dynamic maneuvers (jumping gaps, climbing high obstacles) with a single vision-based NN policy via zero-shot sim-to-real on Unitree A1. **ANYmal Parkour** (Hoeller et al., Science Robotics, 2024) extends this to ANYmal with end-to-end depth-vision RL. (arXiv) (arXiv)

Boston Dynamics Spot: the RL pipeline now exists

Miller et al. (2025; arXiv:2504.17857) published the **first end-to-end RL locomotion policy deployed on Spot hardware**. Using the Spot RL Researcher Kit with Isaac Lab for training and Jetson AGX Orin for inference, they achieved **5.2 m/s locomotion (3× Spot's default max speed)** with robustness to slippery surfaces. Their novel sim-to-real gap quantification uses Wasserstein Distance and Maximum Mean Discrepancy (arXiv) optimized via CMA-ES. The **Spot RL Researcher Kit** (Boston Dynamics + NVIDIA + AI Institute, 2024–2025) provides joint-level control API, Jetson AGX Orin payload mount, and Spot simulation in Isaac Lab — the essential hardware/software stack for learned policy deployment. (NVIDIA Developer) (Blockchain News) Boston Dynamics' blog "Starting on the Right Foot with Reinforcement Learning" (2024) describes their production integration of RL alongside MPC, (Isaac Lab) with **2,000+ hours/week of robustness fleet testing**. (Boston Dynamics)

Ghost Robotics Vision 60: the research gap

The Vision 60 has very limited published RL research. Ghost Robotics' control architecture is proprietary and relatively closed compared to Spot's new RL Researcher Kit. Specifications: 51 kg, 10 kg payload, IP67, −40°C to 55°C, (Ghostrobotics) (Army Technology) 3 m/s sprint, 3.15 hrs continuous walking. (Army Technology) (Ghostrobotics) The only substantial academic evaluation comparing both platforms is **Misenti et al.** (Int. J. Intelligent Robotics and Applications, 2024; DOI:10.1007/s41315-024-00412-4), which tested Spot and V60 stability under dynamic naval conditions. (springer) (Springer) Key finding: **V60 exhibits superior stability and lower peak torques under dynamic ground motions**. (springer) (Springer) Data: github.com/purdue-tracelab/quadruped_assessment. The DHS S&T Urban OpEx 2022 report provides first-responder operational evaluation of V60 (DHS) in subway and post-explosion scenarios. Building a URDF/MJCF model of V60 with system identification via PACE or SPI-Active will likely be necessary for the project.

ManyQuadrupeds: a single policy across diverse morphologies

Shafiee et al. (ICRA 2024; arXiv:2310.10486; miladshafiee.github.io/ManyQuadrupeds) is among the most directly applicable papers. It trains a **single locomotion policy controlling diverse**

quadrupeds (12–16 DOF, 2–200 kg, 10–70 cm height) using Central Pattern Generators combined with RL. (arXiv) (arXiv) The policy modulates CPG parameters while an inverse kinematics layer handles morphological differences. (arXiv) Successful sim-to-real on Unitree Go1 and A1 with 125% added mass. (arXiv) (Miladshafiee) This validates the fundamental feasibility of a shared policy across heterogeneous quadruped platforms. (arXiv)

"Sight Over Site" and perception-aware RL

Kuhlmann et al. (arXiv:2509.17877, September 2025) reframes autonomous robotic inspection as a perception-aware problem where the RL policy learns to find the shortest trajectory guaranteeing **visual contact** with the inspection target rather than navigating to a predefined location. (arXiv) (University of Zurich) The policy exhibits emergent behavioral switching: prioritizing navigation when distant and viewpoint optimization when near. (University of Zurich) Trained end-to-end in simulation and deployed on real robots. (ResearchGate) Affiliations: ETH Zurich (Scaramuzza, Pollefeys, Cadena) and University of Zurich. (arXiv) Project: sight-over-site.github.io. While single-robot and not quadruped-specific, the perception-aware formulation could inform how heterogeneous quadrupeds coordinate for inspection tasks, and the sim-to-real pipeline is directly applicable.

VR-Robo: photorealistic sim-to-real

VR-Robo (arXiv:2502.01536, 2025) reconstructs real indoor environments using Gaussian Splatting from iPhone photos to build interactive simulations, then trains RL policies with visual domain randomization and deploys zero-shot on real quadrupeds — a novel alternative to pure randomization. (arXiv)

3. Applied work on quadruped team coordination

Multi-quadruped coordination on real hardware

decPLM (Pandit et al., 2025; arXiv:2509.14342; decplm.github.io) is the state-of-the-art for decentralized multi-quadruped coordination with sim-to-real transfer. It trains decentralized RL policies (MAPPO in Isaac Lab) enabling teams of Unitree Go2 robots with arms to cooperatively transport objects through contact forces alone — no communication, no rigid coupling.

(Bikram Pandit) (arXiv) **A single policy trained on 2 robots generalizes to teams of 2–10 without retraining.** (Bikram Pandit) validated on real hardware for 2- and 3-robot teams. (The Moonlight)

(Bikram Pandit)

Safe Distributed Learning-Enhanced Predictive Control (arXiv:2503.05836, 2025; github.com/YONEX4090/MultiQuadruped-Control) introduces Scale-Adaptive Permutation-Invariant Encoding (SAPIE) for multi-quadruped formation control with CBF-based collision avoidance, validated on real custom quadrupeds with DDS-based low-latency communication.

(arXiv) **Norén et al.** (arXiv:2602.18260, 2026) propose role-adaptive collaborative formation planning with dynamic role assignment for quadruped teams, using virtual spring-damper systems for inter-robot safety and FM2 for global planning, validated on real hardware. (arXiv) **Dual-Quadruped Collaborative Transportation** (arXiv:2602.16353, 2026) uses constrained RL with cost-advantage decomposition for safe two-quadruped transport. (arXiv) **CoLF** (arXiv:2602.07776, 2026) uses asymmetric leader-follower policies with vision-language guidance, demonstrated on real two-quadruped teams across five scenarios. (arXiv)

MQE (Xiong et al., 2024; ziyanx02.github.io/multiagent-quadruped-environment) is an open-source Isaac Gym-based multi-agent quadruped simulation platform with cooperative transport, tug-of-war, and adversarial tasks using Unitree Go1 models — a useful training environment for prototyping. (Ziyanx02)

Heterogeneous robot teams mixing platforms

MOSAIC (arXiv:2601.23038, 2026) demonstrates scalable autonomy for heterogeneous teams including **ANYmal D quadrupeds**, **wheeled Clearpath Huskies**, and robots with arms, enabling a single operator to supervise teams at different autonomy levels. Validated through real exploration and analysis missions. (arXiv) **TZPP** (arXiv:2603.21723, 2026; github.com/triple-zeroopp/Triple-zero-robot-agent) demonstrates collaboration between a Unitree G1 humanoid and Go2 quadruped with zero training, using MLLM guidance. (arXiv) Ghufra et al. (arXiv:2403.14029, 2024) coordinate a quadruped leader guiding quadcopter followers via affine transformation. (arXiv) (arXiv)

DARPA Subterranean Challenge: the most relevant systems-level references

Team CERBERUS (Alexis et al., Science Robotics, 2023; DOI:10.1126/scirobotics.abp9742; Tranzatto et al., Field Robotics, 2022; arXiv:2201.07067) — **SubT Challenge winner** — deployed ANYmal C quadrupeds with collision-tolerant aerial robots, (IEEE Spectrum) featuring resilient multi-modal sensor fusion, unified exploration planning, and communication via breadcrumb nodes. (Science) (ResearchGate) Partners included ETH Zurich, UNR, NTNU, UC Berkeley, Oxford. (Ox) The Field Robotics paper lists open-source repositories.

Team CoSTAR/NeBula (Agha-mohammadi et al., 2021; arXiv:2103.11470) — 2nd place overall — deployed **Boston Dynamics Spot** alongside tracked robots and UAVs using the NeBula (Networked Belief-aware Perceptual Autonomy) framework for reasoning in belief space with multi-modal positioning and traversability analysis. (arXiv) This is the most relevant reference for Spot in a heterogeneous multi-robot team.

Team MARBLE (Biggie et al., Frontiers in Robotics and AI, 2023; DOI:10.3389/frobt.2023.1109876) placed 3rd, fielding **two Spot quadrupeds** alongside Husky wheeled robots with terrain-aware semantic mapping using semantically-encoded 3D grid maps with low-bandwidth sharing. (Frontiers)

CTU-CRAS-NORLAB (Bayer et al., Field Robotics, 2023) deployed **three Spot quadrupeds** alongside tracked and wheeled robots with topometrical coordination and compatibility-aware traversability mapping — the largest published multi-Spot deployment. Post-event testing demonstrated three Spots exploring >500m fully autonomously. ([ResearchGate](#))

Team Explorer (CMU, arXiv:2404.17759, 2024) contributed a "sliding mode autonomy" architecture with modular capability definitions — particularly relevant for heterogeneous teams. ([arXiv](#))

The Spot + Vision 60 gap: confirmed

No published multi-robot coordination work uses both Spot and Vision 60. The only paper testing both platforms is Misenti et al. (2024), a benchmarking comparison ([ResearchGate](#)) ([Springer](#)) rather than a coordination study. Patent litigation between Boston Dynamics and Ghost Robotics (2022–2025), settled January 2025, ([Circuit](#)) likely discouraged researchers from combining platforms. The Misenti paper provides the only side-by-side performance data: V60 shows superior stability under dynamic ground conditions, ([springer](#)) ([Research Square](#)) suggesting natural role differentiation (V60 for rough terrain; Spot for precision tasks).

Search-and-rescue and military deployments

Peña Queralta et al. (IEEE Trans. Systems, Man, and Cybernetics, 2021; arXiv:2008.12610) provide a comprehensive survey of collaborative multi-robot SAR systems covering heterogeneous teams and active perception. ([arXiv](#)) ([IEEE Xplore](#)) For Spot in SAR, a multimodal system (Robotics and Autonomous Systems, 2022) uses Spot as a UAV carrier for subterranean operations.

Ghost Robotics V60 is **operationally deployed by USAF** ([IEEE Spectrum](#)) at Tyndall AFB (325th Security Forces Squadron, ([Born to Engineer](#)) 4 V60s for perimeter security since 2022), ([The War Zone](#)) ([Army Technology](#)) Cape Canaveral, ([Task & Purpose](#)) and multiple other installations via SBIR Phase II contracts. ([SBIR](#)) ([Ghostrobotics](#)) Ghost Robotics (acquired by LIG Nex1 in 2024 for **\$400M**) has shipped 1,000+ V60 units. ([eWEEK](#)) However, these are operational deployments, not research publications on coordination. **No specific published AFRL research on multi-quadruped coordination was found** — AFRL's robotic coordination work is predominantly UAV/swarm-focused, though their 2024–2025 research priorities explicitly list sim-to-real transfer and multi-agent autonomous systems as challenges.

4. Top recommendations: the 15 most actionable papers

These papers provide the most directly borrowable components — algorithms, architectures, code, evaluation protocols, or baselines — for a shared PPO policy with capability-conditioned observations coordinating Spot and V60.

#	Paper	What to borrow	Code
1	MAPPO (Yu et al., NeurIPS 2022)	Core algorithm: shared PPO, value normalization, implementation tricks	github.com/marlbenchmark/on-policy
2	HAPPO/HARL (Zhong et al., JMLR 2024)	Heterogeneous-agent baseline without parameter sharing; advantage decomposition	github.com/PKU-MARL/HARL
3	Torbati et al. (CoRL 2024)	Capability-conditioned architecture with GNN communication — closest to project design	Described in detail; no public repo
4	HetGPPO (Bettini et al., CoRL 2023)	GNN-based heterogeneous PPO with real-world validation	github.com/proroklab/HetGPPO
5	ManyQuadrupeds (Shafiee et al., ICRA 2024)	Single shared locomotion policy across diverse quadruped morphologies	miladshafiee.github.io/ManyQuadrupeds
6	SHPPPO (Guo et al., Neurocomputing 2025)	Heterogeneous layer + shared parameters architecture	Based on HAPPO codebase
7	RODE (Wang et al., ICLR 2021)	Role emergence via action-effect clustering; bi-level hierarchy	github.com/TonghanWang/RODE
8	Miller et al. (arXiv:2504.17857, 2025)	Complete Spot RL sim-to-real pipeline; CMA-ES sim parameter optimization	Spot RL Researcher Kit + Isaac Lab
9	Legged Gym / Isaac Lab (Rudin et al., CoRL 2022)	Massively parallel quadruped training infrastructure	github.com/leggedrobotics/legged_gym
10	PACE (Bjelonic et al., 2025)	System identification for both platforms	github.com/leggedrobotics/pace-sim2real

#	Paper	What to borrow	Code
11	decPLM (Pandit et al., 2025)	Decentralized multi-quadrupted coordination with MAPPO + sim-to-real	decplm.github.io
12	Huang & Ontañón (2022)	Invalid action masking in PPO for heterogeneous action spaces	github.com/vwxyzjn/invalid-action-masking
13	MAT (Wen et al., NeurIPS 2022)	Attention-based multi-agent architecture with advantage decomposition	github.com/PKU-MARL/Multi-Agent-Transformer
14	RMA (Kumar et al., 2021)	Rapid motor adaptation module for sim-to-real robustness	ashish-kmr.github.io/rma-legged-robots
15	CoSTAR/NeBula (Agha-mohammadi et al., 2021)	Heterogeneous fleet architecture with Spot; belief-space autonomy	arXiv:2103.11470

5. Frameworks, benchmarks, and simulation environments

MARL frameworks

EPyMARL (github.com/uoe-agents/epymarl; Papoudakis et al., NeurIPS 2021) is the strongest recommendation for the project: it natively supports PettingZoo, includes MAPPO/IPPO/MADDPG, ([arXiv](https://arxiv.org/abs/2407.11470)) ([Agents-lab](https://agents-lab.github.io)) migrated to Gymnasium in v2.0.0 ([GitHub](https://github.com)) (July 2024), and integrates with VMAS and SMACv2. ([GitHub](https://github.com)) **HARL** (github.com/PKU-MARL/HARL) provides the HAPPO/HATRPO/HASAC family ([GitHub](https://github.com)) across 7 benchmark environments. **JaxMARL** (github.com/FLAIROx/JaxMARL; Rutherford et al., NeurIPS 2024) achieves up to **12,500× speedup** over CPU approaches ([arXiv](https://arxiv.org/abs/2407.11470)) with JAX-native MAPPO, QMIX, VDN, and includes ([Foersterlab](https://foersterlab.github.io)) SMAX (JAX-based SMAC replacement), MABrax, and Overcooked implementations. **MARLlib** (github.com/Replicable-MARL/MARLlib; ([OpenReview](https://openreview.net/forum?id=2023.07.18v1)) Hu et al., JMLR 2023) unifies 18 algorithms on Ray/RLlib but is pinned to older dependency versions. **RLlib** (part of Ray; Liang et al., ICML 2018) offers industry-grade distributed multi-agent support with flexible policy mapping. **CleanRL** (github.com/vwxyzjn/cleanrl; Huang et al., JMLR 2022) provides transparent single-file PPO implementations with PettingZoo integration — best for prototyping and understanding implementation details. **PettingZoo** (github.com/Farama-Foundation/PettingZoo; Terry et al., NeurIPS 2021) v1.25.0 is the project's chosen multi-agent

API; its Parallel API directly supports simultaneous-action environments. **BenchMARL** (github.com/facebookresearch/BenchMARL) provides TorchRL-based standardized evaluation.

Standard benchmarks

SMAC/SMACv2 (github.com/oxwhirl/smac, github.com/oxwhirl/smacv2; Samvelyan et al., AAMAS 2019; Ellis et al., NeurIPS 2023) remains the gold standard for cooperative MARL evaluation but requires StarCraft II. **MPE** (included in PettingZoo) provides simple 2D continuous cooperative/competitive tasks — ideal for initial algorithm validation. **MaMuJoCo** (in Gymnasium-Robotics) uses PettingZoo ParallelEnv API for multi-agent continuous control — the multi-agent Ant (4-legged) is a natural stepping stone before full quadruped simulation. **MARBLER** (github.com/GT-STAR-Lab/MARBLER; Torbati et al., 2023, IEEE MRS Best Paper) uses Georgia Tech's Robotarium for multi-robot RL benchmarking with real hardware access. Its JAX-accelerated successor **JaxRobotarium** (2025) integrates with JaxMARL for 20× training speedup. **Hanabi** (github.com/deepmind/hanabi-learning-environment) tests theory-of-mind under partial observability. **Overcooked-AI** (github.com/HumanCompatibleAI/overcooked_ai) tests cooperative specialization.

Quadruped simulation environments

NVIDIA Isaac Lab (github.com/isaac-sim/IsaacLab; Mittal et al., 2025) is the premier GPU-accelerated simulation framework, with 16+ robot models including Spot, ANYmal, and Unitree quadrupeds, 30+ ready-to-train environments, and demonstrated zero-shot sim-to-real. It supports multi-agent RL but requires a custom PettingZoo wrapper. **Legged Gym** (github.com/leggedrobotics/legged_gym) remains historically influential but is migrating to Isaac Lab. **MuJoCo Playground** (github.com/google-deepmind/mujoco_playground; Zakka et al., RSS 2025) is the newest entrant: pip-installable, GPU-accelerated via MJX, with quadruped locomotion tasks and zero-shot sim-to-real on Go1 — no native multi-agent support but MaMuJoCo provides the PettingZoo bridge. **PyBullet** (github.com/bulletphysics/bullet3) is free and lightweight with Minitaur/Laikago environments but CPU-only. **Brax/MABrax** (github.com/google/brax) provides JAX-native differentiable physics with multi-agent Brax environments in JaxMARL. **Quad-SDK** (github.com/robomechanics/quad-sdk; Norby et al., ICRA 2022 Workshop) is a ROS-based full-stack framework with MPC control — useful for deployment integration but not RL-focused. **MQE** (ziyanx02.github.io/multiagent-quadruped-environment) provides Isaac Gym-based multi-agent quadruped tasks — the closest existing multi-agent quadruped simulator.

6. Gaps and opportunities this project can address

The primary novelty is clear: no published work on learned coordination between Spot and Vision 60 exists. Beyond this headline gap, several specific opportunities emerge from the literature:

Most heterogeneous MARL work operates in grid-worlds or simple 2D particle environments, not continuous 2D with obstacles and rooms. The canonical benchmarks (SMAC, MPE) use simplified dynamics. MARBLER uses ground robots but not legged platforms. The project's PettingZoo-style 2D simulation with obstacles and rooms sits in an underexplored middle ground between toy benchmarks and full 3D physics — sufficiently complex for meaningful coordination while tractable enough for thorough MARL experimentation.

Vision 60 has no public sim-to-real RL pipeline. Unlike Spot (which now has the RL Researcher Kit, Isaac Lab integration, and Miller et al.'s complete pipeline), V60 has no published URDF/MJCF model, no open actuator dynamics data, and no demonstrated RL locomotion. Building this pipeline — using PACE or SPI-Active for system identification from limited real-world data — would itself be a contribution.

Capability-conditioned policies have not been validated on real heterogeneous legged robots. Torbati et al. (2024) validated on simulation and ground robots; HetGPPO validated on wheeled robots; ManyQuadrupeds demonstrated morphology generalization but for single-agent locomotion, not multi-agent coordination. Demonstrating that capability conditioning enables effective coordination between physically distinct quadrupeds would bridge a gap between the MARL architecture and robotics communities.

The MARL-to-quadruped bridge is almost entirely unbuilt. The decPLM and MQE works are initial steps, but both use homogeneous Unitree platforms. No work combines heterogeneous MARL algorithms (HAPPO, capability-conditioned MAPPO) with heterogeneous quadruped locomotion and sim-to-real transfer. This three-way integration — heterogeneous MARL × quadruped sim-to-real × physical deployment — is novel.

Role emergence in physically grounded heterogeneous teams is understudied. RODE and ROMA demonstrate emergent roles in StarCraft; the project could show whether physical capability differences (Spot's payload vs. V60's terrain stability, different sensor suites) produce qualitatively different emergent role structures than capability-symmetric agents with learned role differentiation.

AFRL has no published multi-quadruped coordination research despite operating V60 fleets and listing multi-agent autonomy and sim-to-real as research priorities. The project's AFRL collaboration could fill this gap directly.

Curriculum learning for heterogeneous multi-robot teams is nascent. Existing MARL curricula vary team size or task difficulty but not agent capabilities. A curriculum that progressively introduces heterogeneity — starting with identical agents, then gradually differentiating capabilities — has not been studied.

Evaluation protocols for multi-robot quadruped coordination do not exist. The project has an opportunity to define standardized metrics: coordination efficiency (time-to-complete vs. single-

robot), communication overhead, role differentiation index, robustness to single-agent failure, and sim-to-real transfer degradation for team performance vs. individual performance.

These gaps collectively position the project to contribute a novel problem formulation (heterogeneous quadruped coordination with specific commercial platforms), a new training architecture (capability-conditioned shared PPO validated on physically distinct legged robots), and an evaluation framework that could serve future work in this emerging intersection.